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MID-SEMESTER EXAMINATION, MAR-2025

Name of the Program: B.Tech
Name of the Course : Structural Analysis
Time: 90 minutes

Semester : IV semester
Course Code: TCE 402
Maximum Marks: 50

Note: *i) Answer all questions by choosing any one of the sub questions.*
ii) Each question carries 10 marks.

Q.1		(1 X 10 = 10 Marks)	
	a)	A fixed beam of span L is subjected to a uniformly distributed load (UDL) of intensity 'w' over its entire span. Determine the fixed-end moments at both supports and determine their values for L=6m and w=10kN/m.	(CO 1)
		(OR)	
	b)	A fixed beam with a span of L carries a central concentrated load P. Derive the expression for the fixed-end moments at both supports and determine their values for P=30kN and L=4m	(CO 1)
Q.2		(1 X 10 = 10 Marks)	
	a)	A simply supported beam AB of span 4m carries a point load of 100KN at its center C. The values of I for the left half is 1×10^8 mm^4 and for the right half portion I is 2×10^8 mm^4. Find the slopes at the two supports and deflection under the load using conjugate beam method. Take $E=2 \times 10^5$ N/mm^2	(CO 1)
		(OR)	(CO 1)
	b)	Using moment area method, determine the mid-span deflection in a simply supported beam of span 'L', when it is subjected to a udl throughout the span.	(CO 1)
Q.3		(1 X 10 = 10 Marks)	
	a)	A cantilever of length L is carrying a point load of W KN at a distance of 'a' from the fixed end. Calculate Slope at the free end and deflection at the free end.	(CO 1)
		(OR)	(CO 1)
	b)	A beam 4m in length is simply supported at the ends and carries a uniformly distributed load of 6kN/m length. Determine the strain energy stored in the beam. Take $E=200GN/m^2$ and $I=1440$ cm^4	(CO 2)
Q.4		(1 X 10 = 10 Marks)	
	a)	Find an expression for strain energy due to bending for a beam of length l simply supported at the ends and carrying a uniformly distributed load w/unit run over whole of its span. The beam is of constant cross-section throughout its length having flexural rigidity as EI	(CO 2)
		(OR)	(CO 2)
	b)	The external diameter of a hollow shaft is twice the internal diameter. It is subjected to pure torque and it attains a maximum shear stress τ. Show that the strain energy stored per unit volume of the shaft is $\frac{5\tau^2}{16C}$. Such a shaft is required to transmit 5400kW at 110 r.p.m with uniform torque, the maximum stress not exceeding 84 MN/m^2. Determine: i) The shaft diameters ii) The energy stored per m^3	(CO 2)

Q.5		(1 X 10 = 10 Marks)	
	a)	Compare the strain energies of the following two shafts subjected to the same maximum shear stress in torsion: i) A hollow shaft having outer diameter n times the inner diameter ii) A solid shaft Masses, lengths and materials of the two shafts are the same.	(CO 2)
		(OR)	(CO 2)
	b)	A beam of length l simply supported at the ends is loaded with a point load W at a distance 'a' from one end. Assuming that the beam has constant cross-section with moment of inertia I and young's modulus E, find the strain energy of beam and hence find the deflection under the load. Stain energy due to shearing is neglected.	(CO 2)